

ConfidenceOS

Read-only trust-aware HMI compiler and Runtime for next-generation industrial control interfaces

ABB Accelerator Grand Finale - Project Documentation

ConfidenceOS addresses alarm fatigue and HMI engineering effort by turning raw tags, metadata, and live sensor trust evidence into generated operator screens. It does not replace DCS/HMI systems and never writes control commands.

Theme Fit	Next-gen control system interface
Primary Users	Operators, maintenance technicians, engineers, managers, auditors
Core Idea	Confidence scoring and trust state for every critical process reading
Main Flow	Studio -> generated Runtime -> abnormal situation -> verification -> handover debt
Deployment Posture	Software simulator, CSV replay placeholder, OPC UA read-only boundary, Docker supported

1. Executive Summary

ConfidenceOS reframes the HMI from a static alarm display into a trust-aware operating workspace. The central question is not only what the sensor value is, but whether that reading can safely be used as the basis for an operator decision.

Studio compiles screens from asset metadata and reusable templates. Runtime consumes the published manifest and presents operating-basis views under pressure. The system combines deterministic confidence scoring, physics-based mass balance, role-aware interfaces, verification workflows, and shift handover traceability.

Core Value Proposition

- Reduce alarm fatigue by collapsing related low-level warnings into one abnormal situation and one operating question.
- Improve operator decisions by showing do-not-trust, trusted substitute, first safe action, decision freeze, and exit condition.
- Lower HMI engineering effort by generating screens from asset models, signal bindings, templates, and validation receipts.
- Improve consistency through deterministic template tests, publish guardrails, and screen receipts.
- Keep the safety posture honest: ConfidenceOS is read-only decision support, not autonomous control.

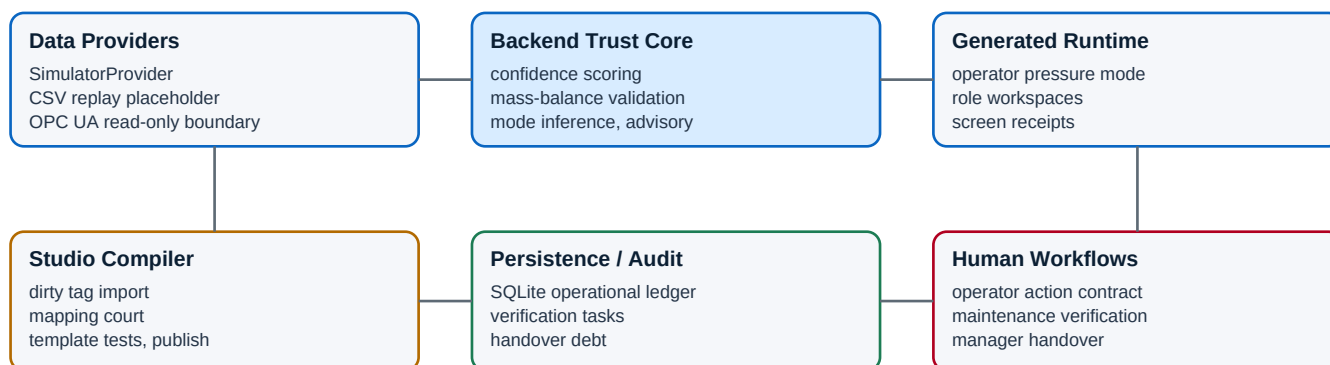
Capability	Implemented Mechanism
Confidence at center	Score, tier, trust state, evidence, counter-evidence, recommended action, and related assumptions.
Alarm-fatigue reduction	Alarm collapse maps raw warnings into one abnormal situation and operating basis.
Engineering acceleration	Compiler pipeline turns dirty tags and templates into generated Runtime screens.
Role/context awareness	Operator, Maintenance, Engineer, Manager, and Auditor workspaces render different operational content.
Traceability	Operational ledger links incidents, verification tasks, handover debt, and compliance evidence.

2. Problem Statement and Theme Alignment

Traditional HMIs produce streams of alerts and color-coded indicators. Operators must manually determine which signals matter, whether a reading is trustworthy, and which actions should be frozen until verification. This increases cognitive load during abnormal situations.

ABB desired solution item	ConfidenceOS response
Auto-generated HMI screens	Compiler-generated screens, faceplates, role sections, stress panels, and receipts.
Signal templates and libraries	Signal templates plus equipment templates for vessel, valve, flow pair, pump, and abnormal situation.
Metadata-driven engineering	Plant/area/unit/module/equipment/signal graph drives Runtime navigation and generation.
AI-assisted configuration	Optional LLM gateway for explanations; deterministic rules remain authoritative and engineer approval is required.
Low-code/no-code tools	Studio supports approve, ignore with reason, assign template, run build, preview, publish, and reset.
Role and context aware UI	Pressure mode removes nonessential panels and shows the operating basis.
Modern UX patterns	Compiler workflow, Mapping Court, Trust Map, Teams-like Shift Channel, and screen receipts.

3. System Architecture

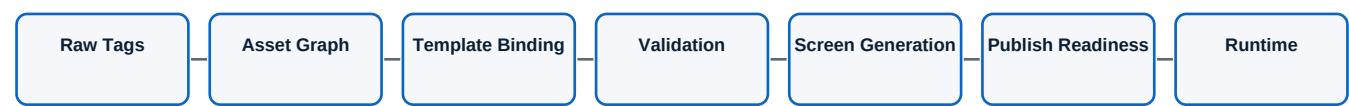


Architecture Notes

- Frontend: React, Zustand, Vite, role-based routes, generated Runtime renderer, Studio workspace, Shift Channel, support views.
- Backend: FastAPI, deterministic trust and advisory services, simulator tick loop, SQLite persistence, JWT role auth, compiler services.
- Data model: JSON asset models, signal templates, equipment templates, role policies, context policies, assumptions register, imported tag fixtures.
- Runtime data: WebSocket live stream for readings, confidence, mass balance, incidents, timeline, verification tasks, and handover debt.
- Integration boundary: SimulatorProvider is active; CsvReplayProvider and OpcUaProvider show read-only extensibility without control writes.

ConfidenceOS reads simulator/provider tags and emits trust/evidence metadata. It does not write tag values, setpoints, controller modes, pump starts/stops, or alarm acknowledgements.

4. HMI Compiler Pipeline



Studio is the engineering workspace. It turns raw tag imports and asset metadata into a published Runtime manifest. The compiler intentionally refuses publish when unresolved dirty tags or blocking validation issues remain.

Stage	Purpose	Example
Raw Tags	Import dirty provider tags such as U15_LT_5100.PV and BAD_TAG_123.	Mapped, ambiguous, ignored, or blocking buckets.
Asset Graph	Resolve tags to equipment/signal relationships.	LT-5100 -> V-5100 primary level.
Template Binding	Attach reusable templates to equipment.	V-5100 -> vessel, P-101 -> pump.
Validation	Apply INFO/WARNING/BLOCKING guardrails.	Publish refused until critical mappings are approved.
Screen Generation	Create screens, faceplates, role views, receipts.	Operator stress mode and vessel faceplate.
Publish Readiness	Allow only passing builds or pass-with-warning builds.	Published build id, manifest, diff, receipts.

Every generated screen and faceplate carries provenance: build id, asset id, template id/version, source tags, role policy, context policy, validation status, generated-because lines, warnings, and source files.

5. Confidence and Trust Methodology

ConfidenceOS treats confidence as a governed trust rubric, not a calibrated probability. The score explains whether a reading can be used as an operating basis. Runtime maps scores and process contradictions into trust states.

Trust State	Runtime Meaning
TRUSTED	Reading can be used as normal operating evidence.
DEGRADED	Reading is visible but requires caution and supporting evidence.
QUARANTINED	Reading remains visible but cannot be used as basis for blocked decisions.
SUBSTITUTED	Reading or inferred variable is being used as trusted substitute evidence.
UNAVAILABLE	Reading is missing, stale, or not bound to live data.

Sub-scores include calibration, stability, cross-sensor evidence, physical plausibility, and mass-balance evidence. During the demo, LT-5100 is quarantined because low level confidence and FI/FO mass-balance contradiction make the indicated level unreliable.

6. Runtime Experience

Runtime consumes the published manifest and live stream, then renders different workspaces based on role and context. In warning, critical, mass-balance divergence, or manual-verification context, pressure mode deletes nonessential panels and centers the operating basis.

Runtime Feature	Functionality
Trust Map Navigation	Plant -> area -> unit -> module -> equipment -> signal navigation with trust hotspots.
Generated Faceplates	Template-based equipment panels with source tags, live values, trust state, and receipt summary.
Operating Basis Ledger	Statements with owner, evidence, status, and exit condition.
Situation Workspace	Abnormal situation, do-not-trust, trusted substitute, first safe action, decision freeze, exit condition, evidence.
Alarm Collapse Receipt	Raw signal count, suppressed alarms, operator question, collapse reason, and consumed signals.
Interaction Compression Estimate	Transparent estimate of raw signals and blocked decisions reduced into required operator actions.

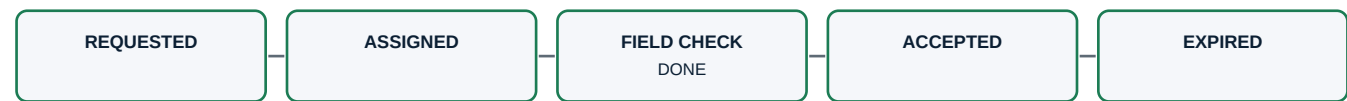
Operator Action Contract

- Do not trust: LT-5100 as sole level basis.
- Use instead: FI-2010, FO-2020, PT-3100, manual level check, sight glass, flow-implied level.
- First safe action: verify level locally before increasing feed.
- Decision freeze: increase feed, increase load, accept handover without verification.
- Exit condition: affected level trust restored to TRUSTED or manual verification accepted.

7. Role-Based Workspaces

Role	Primary Work Shown
Operator	Single safe move, operating basis, do-not-trust, trusted substitute, decision freeze, exit condition.
Maintenance	Verification task, calibration context, device health, confidence debt, field check status.
Engineer	Signal binding, template receipts, assumptions, sensitivity, validation warnings, build provenance.
Manager	Unresolved handover debt, decision freeze state, handover acceptance status, timeline evidence.
Auditor	Operational event ledger, compliance report, provenance, read-only evidence references.

Verification Task Lifecycle



Verification tasks never override confidence. They record temporary field verification and can block handover acceptance until accepted or expired according to the workflow.

The Shift Channel acts like an operational thread. It pins unresolved incidents, active decision freezes, low-confidence critical sensors, active verification tasks, confidence debt, and ledger references.

8. UI Evidence - Studio Compiler

ConfidenceOS

RuntimeStudioShift Channel

SUPPORT VIEWS

Engineer4 Trust Alerts2 Unack AlarmsIntegrity 80%

LIVE

SIGN OUT

HMI COMPILER

Raw Tags → Asset Graph → Template Binding → Publish Readiness ...

Ready to publishpublish gate: ready after engineering reviewhmi-build-0004

Engineerread-only trust-aware HMI compiler

Texas City VesselL2Engineering Configuration Workspace

CONFIDENCEOS STUDIO

Compiler Workflow

READY

ASSET MODEL

Texas City Training Model Vessel

NEXT ENGINEERING STEP

Publish the generated Runtime when the preview matches the engineering intent.

COMPILER RUNBOOK

Map raw provider tags into asset-model signals.

Resolve Mapping Court blockers with approval or ignore reason.

Run build to validate templates and publish guardrails.

Preview generated Runtime, then publish only if readiness is clear.

COMPILER STEP 1

READY

1. Run deterministic mapping

Create explainable tag-to-asset proposals.

Nothing is published.

8 imported tags in Mapping Court.

COMPILER STEP 2

PASS_WITH_WARNINGS

2. Validate HMI build

Run compiler guardrails across mappings, templates, roles, and context policies.

Build can pass with warnings.

COMPILER STEP 3

READY

3. Generate publish readiness

HMI COMPILER PIPELINE

Raw Tags To Runtime Build

PASS_WITH_WARNINGS

STAGE

PASS

Raw Tags

Raw provider tags loaded from the selected asset model source.

STAGE

WARN

Asset Graph

Tags proposed against canonical signals with evidence and approval state.

STAGE

PASS

Template Binding

Reusable equipment templates bound to available signal roles.

STAGE

WARN

Validation

Guardrails check missing signals, role visibility, and critical context behavior.

STAGE

PASS

Screen Generation

Runtime manifests, faceplates, receipts, and stress panels generated.

STAGE

PASS

Publish Readiness

Publish allowed only when blocking issues are cleared.

BUILD ID

hmi-build-0004

PUBLISH GATE

can publish latest build

COMPILER CONTRACT

read-only trust-aware HMI layer beside existing DCS/HMI

forbidden writes: tag values / setpoints / controller modes / DCS alarm acknowledgements

DIRTY TAG IMPORT GAUNTLET

Imported Raw Tags

Raw import only: canonical mappings are derived by Mapping Court from the active asset model.

MAPPED	AMBIGUOUS	UNMAPPED	IGNORED	BLOCKING
6	0	0	2	0

U15_LT_5100.PV	unresolved	Derived Suggestion	Mapped State
RAW IMPORT ROW	DERIVED SUGGESTION		
15-FI-2010	unresolved	Derived Suggestion	Mapped State
RAW IMPORT ROW	DERIVED SUGGESTION		

ConfidenceOS — read-only HMI overlay

System LogsUTC: 3:58:51 PMRuntime readyLive stream connectedLast Tick: 3:58:51 PMHealth: 3:58:41 PM

Engineer Studio after publishing hmi-build-0004. The workspace exposes compiler stages, dirty tag import, validation warnings, publish readiness, and read-only compiler contract.

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Read-only trust-aware HMI layer

9. UI Evidence - Operator Pressure Mode

ConfidenceOS

Runtime

Shift Channel

SUPPORT VIEWS

Operator

4 Trust Alerts

2 Unack Alarms

LIVE

SIGN OUT

1

ABNORMAL SITUATION

Inventory accumulation with unreliable level indication

OPERATING CONTEXT

Texas City Demo Plant / MASS BALANCE DIVERGENCE

Live read-only

Operator

LIVE READ-ONLY

PRESSURE MODE / OPERATING BASIS WORKFLOW

Inventory accumulation with unreliable level indication

Grounded explanation disabled during active decision freeze.

TRAINING SOURCE — SIMULATOR SCENARIO STATE (read-only view — scenario control available to Engineers)

TRUST QUARANTINE / ok / tick 549

Primary level indication frozen while inventory changes

Valve indication disagrees with physical feed effect

TT-4100 calibration drift

PT-3100 stuck_reading

1. Normal operation

2. Mode inferred

3. Level indication loses trust

4. Mass-balance contradiction collapses alarms

5. Operator receives one safe move

6. Verification and handover debt are created

response: Quarantined signal remains visible but cannot be used as decision basis.

recovery: Manual verification is accepted and confidence recovers from deterministic evidence.

workflow: Runtime promotes operating basis and action contract.

MASS-BALANCE EVIDENCE

Mass-Balance CRITICAL: Discrepancy Of 155.2 Ft Exceeds CRITICAL Threshold (20.0 Ft), Flow-Implied Delta: +157.4 Ft, Measured Delta: +2.2 Ft.

OPERATOR SINGLE SAFE MOVE

Verify Level Locally Before Increasing Feed.

DO NOT TRUST

LT-5100

TRUSTED SUBSTITUTE

FI-2010

FO-2020

PT-3100

Manual Level Check

Sight Glass

FI-2010 / FO-2020 Flow-Implied Level

DECISION FREEZE

Increase Feed

Increase Load

Accept Handover Without Verification

EXIT CONDITION

Affected Level Trust Restored To TRUSTED

Manual Level Verification Or Sight Glass Check Accepted

CONFIDENCE EVIDENCE LEDGER

Confidence is a governed trust rubric, not probability. The score explains whether a reading can be used as operating basis.

LT-5100

QUARANTINED / 44%

CALIBRATION

0.74

STABILITY

0.00

CROSS SENSOR

0.20

PLAUSIBILITY

1.00

DOMINANT FACTOR

Stability

FI-2010

SUBSTITUTED / 99%

CALIBRATION

0.97

STABILITY

1.00

CROSS SENSOR

1.00

PLAUSIBILITY

1.00

DOMINANT FACTOR

None

TRAINING STATE

Simulator Source

Phase: TRUST QUARANTINE

Active Failures: 4

Verify Level Locally Before Increasing Feed Or Accepting Handover.

PRESSURE MODE

Operating Basis

Inventory Accumulation With Unreliable Level Indication

Required Operator Actions

1 Action Required

Read-Only Boundary

ConfidenceOS is a read-only trust-aware HMI layer beside existing DCS/HMI. It does not write commands, setpoints, or alarm acknowledgements.

ConfidenceOS — read-only HMI overlay

System Logs

UTC: 3:58:36 PM

Runtime ready

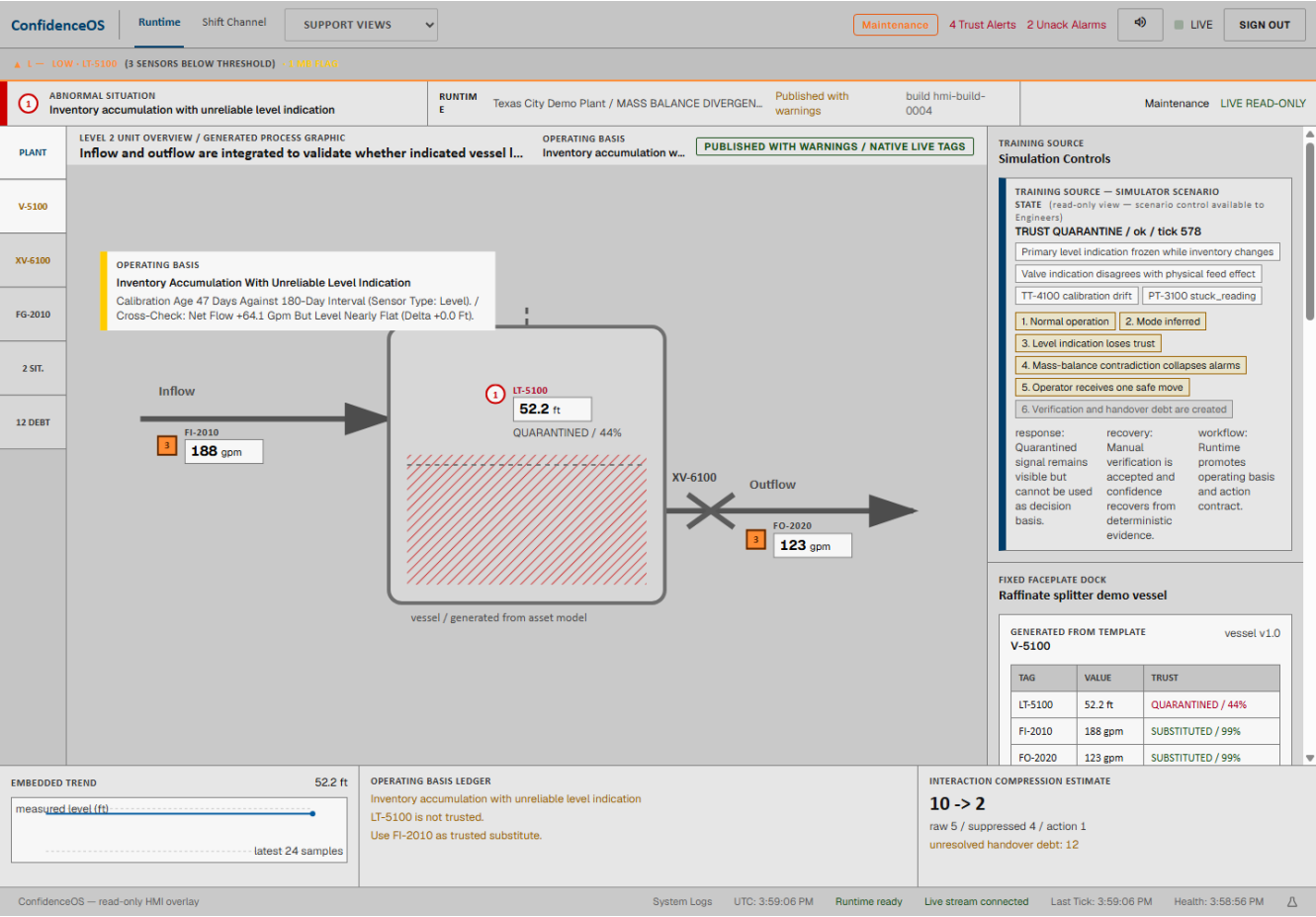
Live stream connected

Last Tick: 3:58:36 PM

Health: 3:58:27 PM

Operator Runtime in pressure mode: abnormal situation, single safe move, do-not-trust, trusted substitutes, decision freeze, exit condition, and evidence ledger.

10. UI Evidence - Maintenance Runtime



Maintenance Runtime shows generated process graphic, trust quarantine state, generated faceplate, verification task context, confidence debt, and trend evidence.

11. UI Evidence - Shift Channel

ConfidenceOS

RuntimeStudioShift Channel

SUPPORT VIEWS

Manager4 Trust Alerts2 Unack AlarmsIntegrity 80%

LIVE

SIGN OUT

INCOMING SHIFT BASIS

Handover BLOCKED - 17 unresolved trust exceptions must be carried and cleared.

- Inventory accumulation with unreliable level indication. Operating basis: Verify level locally before increasing feed.

- Decision freeze: increase_feed. Operating basis: Carry blocked decision and exit condition into next shift.

- Decision freeze: increase_load. Operating basis: Carry blocked decision and exit condition into next shift.

1PINNED HANDOVER DEBT

17 unresolved item(s)

Handover blockedactive verification tasks: 5

plant-a:shift-channelshift continuity console

Unresolved Trust Debt

17 ITEMS

UNRESOLVED_INCIDENT

Inventory accumulation with unreliable level indication

Verify level locally before increasing feed.

UNRESOLVED_INCIDENT

Instrument confidence degraded

Verify ZT-6100 locally or compare against an independent field indication.

ACTIVE_DECISION_FREEZE

Decision freeze: increase_feed

Carry blocked decision and exit condition into next shift.

ACTIVE_DECISION_FREEZE

Decision freeze: increase_load

Carry blocked decision and exit condition into next shift.

ACTIVE_DECISION_FREEZE

Decision freeze: accept_handover_without_verification

Carry blocked decision and exit condition into next shift.

ACTIVE_DECISION_FREEZE

Decision freeze: use_degraded_signal_as_primary_reference

Carry blocked decision and exit condition into next shift.

LOW_CONFIDENCE_CRITICAL_SENSOR

LT-5100 remains LOW confidence

Verify LT-5100 locally or compare against an

Shift Channel

L2Handover & Operating Debt Console / plant-a:shift-channel

C - HANDOVER BLOCKED

Clear active verification tasks to unblock handover

TEAMS-LIKE HANDOVER THREAD

Shift Continuity

Operator

Add operator note for handover...

PIN NOTE

OPERATIONAL EVENT LEDGER

Trace Ledger

60

INCIDENT	VERIFY	HANDOVER	NOTES
41	19	0	0
HANDOVER_DEBT_CREATED / INCIDENT_TIMELINE plant-a:handover_debt:verification_task:verification-task:PT-3100:1782037434-9dc5b1 1782037434-9dc5b1 - Field verification task REQUESTED for PT-3100 3:53:59 PM			
HANDOVER_DEBT_CREATED / INCIDENT_TIMELINE plant-a:handover_debt:verification_task:verification-task:FI-2010:1782037293-7fb27e 1782037293-7fb27e - Field verification task REQUESTED for FI-2010 3:51:38 PM			
HANDOVER_DEBT_CREATED / INCIDENT_TIMELINE plant-a:handover_debt:verification_task:verification-task:FO-2020:1782037293-7e5ea5 1782037293-7e5ea5 - Field verification task REQUESTED for FO-2020 3:51:38 PM			
HANDOVER_DEBT_CREATED / INCIDENT_TIMELINE plant-a:handover_debt:verification_task:verification-task:LT-5100:1782037293-24b7fb 1782037293-24b7fb - Field verification task REQUESTED for LT-5100 3:51:38 PM			
HANDOVER_DEBT_CREATED / INCIDENT_TIMELINE plant-a:handover_debt:verification_task:verification-task:ZT-6100:1782037293-992333 3:51:38 PM			

ConfidenceOS — read-only HMI overlay

System LogsUTC: 3:59:21 PMRuntime readyLive stream connectedLast Tick: 3:59:20 PMHealth: 3:59:11 PM

Manager/Auditor handover view. Unresolved trust debt, active verification tasks, decision freezes, and operational event ledger references are visible before handover acceptance.

12. Metadata-Driven Engineering

The same compiler works from different asset metadata. The Texas City vessel model and Pump Station model share the same pipeline but produce different equipment faceplates, source tags, action language, and receipts.

Asset Model	Primary Equipment	Signals	Generated Templates
Texas City Demo Vessel	V-5100 vessel, XV-6100 valve, FG-2010 flow pair	LT, FI, FO, PT, TT, ZT	vessel, valve, flow_pair, abnormal_situation
Pump Station Demo	TK-100 tank, P-101 pump, FG-100 flow pair	LIT, FIT, VIB, PIT, RUN	vessel, pump, flow_pair, abnormal_situation

Reusable Equipment Templates

Template	Required Signals	Generated Sections
vessel	level, flow_in, flow_out	operating basis, faceplate, mass balance, substitutes, evidence, decision freeze
valve	valve position	faceplate, command/feedback trust, downstream effect
flow_pair	flow_in, flow_out	mass balance, trend, confidence
pump	vibration	device health, discharge/flow context, vibration trust, verification task
abnormal_situation	none	do-not-trust, substitute, safe action, freeze, exit, evidence, timeline

13. Implementation and Tech Stack

Layer	Technologies / Modules
Frontend	React 19, Vite, Zustand, Recharts, D3 force, role-based routes, generated Runtime components, Studio components.
Backend API	FastAPI, Pydantic, Uvicorn, JWT role auth, WebSocket live stream.
Data and Persistence	SQLite through SQLAlchemy, JSON asset/template/assumption registers, persisted operational events and verification tasks.
Simulation	Software simulator, scenario controls, CSV replay placeholder, OPC UA read-only adapter boundary.
Trust Engines	confidence.py, mass_balance.py, advisory.py, mode_inference.py, decision_integrity.py, prediction.py.
Compiler Services	hmi_compiler.py, model_graph.py, template_library.py, screen_generator.py, studio_service.py, template_tests.py.
Operational Workflows	verification_service.py, shift_channel.py, handover.py, operational_ledger.py, assumptions.py.
Deployment	Docker Compose with backend healthcheck and frontend dependency on backend health.

Security and Safety Posture

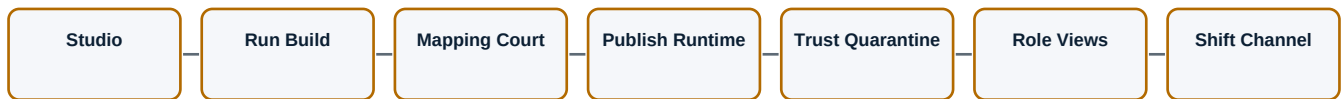
- Role-based login for Operator, Maintenance, Engineer, Manager, and Auditor.
- JWT-protected write-style demo endpoints; read-only boundaries are explicit.
- Scenario loading is restricted to known safe files.
- LLM support is optional and advisory only; deterministic rules and engineer approval remain authoritative.

14. Methodology and Deterministic Logic

The prototype avoids safety-authoritative AI. All publish guardrails, trust state transitions, mode inference, alarm collapse, action contracts, verification blocking, and handover debt are deterministic. Optional AI can explain mapping suggestions, but cannot approve or publish them.

Method	How ConfidenceOS Applies It
Physics-based validation	Flow in/out are integrated to validate level plausibility through mass-balance relationships.
Mode inference	Rules infer STEADY_STATE, STARTUP_RAMP, MASS_BALANCE_DIVERGENCE, INSTRUMENTATION_SUSPECT, MANUAL_VERIFICATION_REQUIRED.
Alarm collapse	Related warnings are grouped into one abnormal situation and one operator question.
Trust quarantine	Critical low-confidence signals with physical contradiction remain visible but are blocked as decision basis.
Assumption governance	Weights, tolerances, thresholds, owners, review dates, approval status, MOC references, and warnings.
Publish guardrails	INFO visible only, WARNING publish allowed, BLOCKING publish refused.
Operational ledger	Stable event IDs connect incidents, tasks, handover, notes, and reports.

15. Recommended ABB Demo Flow



- Open Studio and show the HMI Compiler pipeline.
- Explain dirty tags and Mapping Court: deterministic suggestion active, AI optional, engineer approval required.
- Show PASS_WITH_WARNINGS and publish generated Runtime.
- Open Operator Runtime and show the abnormal situation: inventory accumulation with unreliable level indication.
- Point to Trust Quarantine: LT-5100 is visible but not allowed as decision basis.
- Show one safe move: verify level locally before increasing feed.
- Switch to Maintenance to show field verification task and confidence debt.
- Switch to Manager/Auditor Shift Channel to show handover blocked until trust debt is cleared.

ConfidenceOS is a compiler for trust-aware HMIs. It reduces engineering time by generating screens from metadata, and improves operator decisions by turning raw alarms into an operating basis.

16. Prototype Scope and Honest Boundaries

The prototype is intended for hackathon evaluation and jury testing. It demonstrates end-to-end concepts with deterministic logic and live simulator data, but it is not a certified safety system or production DCS/HMI replacement.

Area	Current Status	Boundary
Simulator	Active	Software process model, not a plant-validated digital twin.
OPC UA	Read-only boundary prepared	No default industrial endpoint, certificates, or write operations.
AI assistance	Optional gateway support	Advisory explanation only; no autonomous approval or control.
Compliance report	Generated evidence package	Not a certified regulatory report or site MOC record.
Forecast	Deterministic trend estimate	Confidence degradation only, not predictive failure.
Second asset model	Pump Station model included	Proves compiler generality but remains demo-scoped.

Future Improvements

- Connect to a real read-only OPC UA demo endpoint with certificates and namespace browsing.
- Add a richer process simulator for non-scripted abnormal situations.
- Improve trend visualization with axes, thresholds, and event overlays.
- Add signed engineering approval workflow for assumptions and template changes.
- Expand template library across pumps, compressors, heat exchangers, reactors, and utilities.

17. Key Files and Modules

File / Area	Purpose
backend/main.py	FastAPI application, plant loops, live APIs, health/readiness, compliance, runtime support.
backend/confidence.py	Confidence scoring and explainability core.
backend/mass_balance.py	Physics-based mass-balance validation.
backend/advisory.py	Incidents, action contracts, alarm collapse, timeline events.
backend/hmi_compiler.py	Compiler build artifact, mapping validation, publish readiness.
backend/screen_generator.py	Generated manifests, faceplates, role sections, receipts.
backend/studio_service.py	Studio state, mapping approvals, build/publish workflow.
backend/verification_service.py	Verification task lifecycle and audit rows.
backend/operational_ledger.py	Append-only event ledger for traceability.
frontend/src/components/RuntimePlatform.jsx	Generated Runtime, pressure mode, role workspaces.
frontend/src/components/StudioWorkspace.jsx	Compiler workspace UI.
frontend/src/components/ShiftChannel.jsx	Handover debt and operational thread.

End of document.